

Metadata Analysis to Identify Digital Footprints

Tool: ExifTool / Metadata2Go

Theory

Every digital file stores hidden information called metadata — such as the author name, creation date, GPS location, camera model, and software used. In a security investigation, metadata acts as a digital footprint that can trace a leaked file back to its origin. ExifTool is a command-line tool used to extract this metadata, and Metadata2Go provides the same functionality through a web browser.

Steps

1. Open the terminal in Kali Linux.
 2. Install ExifTool. Run: `sudo apt install libimage-exiftool-perl -y`
 3. Download a sample image to test. Run: `wget -O test.jpg https://samples.exiftool.org/Canon.jpg`
 4. Extract all metadata from the image. Run: `exiftool test.jpg` — The output shows fields like Author, Camera Model, GPS Latitude, GPS Longitude, Create Date, and Software.
 5. Extract only GPS coordinates. Run: `exiftool -GPSLatitude -GPSLongitude test.jpg`
 6. Extract the author and software fields. Run: `exiftool -Author -Artist -Software test.jpg`
 7. Save the full metadata report to a file. Run: `exiftool test.jpg > metadata_report.txt`
 8. Open Firefox and go to <https://www.metadata2go.com> — Click Choose File, select test.jpg, and click Analyze. The website shows all metadata in a table. Compare results with ExifTool output.
 9. To remove all metadata from a file before sharing. Run: `exiftool -all= test.jpg` — Verify by running `exiftool test.jpg` again; sensitive fields like Author and GPS will be gone.
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Conclusion

ExifTool and Metadata2Go successfully extracted digital footprints such as author name, GPS location, and camera model from the test image. This shows how metadata in ordinary files can reveal sensitive information and aid in tracing the source of a data leak.